

index, i, j, n, m *indices*
label, l, k *field label*
termvar, x
basetype, A

terminals	$::=$ $\rightarrow$ $\rightarrow$ $\Rightarrow$ λ $\mapsto$ $\vdash$ $\emptyset$ $\times$ $<:$ $\langle$ $\rangle$	
labels	$::=$ $l_1 .. l_n$	
Γ	$::=$ $\emptyset$ $\Gamma, x : T$	<i>contexts:</i> <i>empty context</i> <i>term variable binding</i>
formula	$::=$ judgement $formula_1 .. formula_n$ $j \text{ INDEXES } t_1 .. t_n$ $j \text{ INDEXES } T_1 .. T_n$ DISTINCT $l_1 .. l_n$ $x : T \in \Gamma$	
T, S, U	$::=$ Bool Nat $T \rightarrow T'$ (T) A Unit $T_1 \times T_2$ $T_1 + T_2$ $\{ T_i^{i \in 1..n} \}$ $\langle l_i : T_i^{i \in 1..n} \rangle$	<i>types:</i> <i>type of booleans</i> <i>type of natural numbers</i> <i>type of functions</i> M <i>base type</i> <i>unit type</i> <i>product type</i> <i>sum type</i> <i>tuple type</i> <i>type of variants</i>
t	$::=$ (t) true false if t_1 then t_2 else t_3 0 succ t pred t iszero t x	<i>terms:</i> M <i>constant true</i> <i>constant false</i> <i>conditional</i> <i>constant zero</i> <i>successor</i> <i>predecessor</i> <i>zero test</i> <i>variable</i>

termvar
 basetype
 terminals
 labels
 Γ
 formula
 T
 t
 C
 v

$t \longrightarrow t'$ *Evaluation*

$$\begin{array}{c}
 \frac{}{\text{if true then } t_2 \text{ else } t_3 \longrightarrow t_2} \quad \text{E_IFTRUE} \\
 \frac{}{\text{if false then } t_2 \text{ else } t_3 \longrightarrow t_3} \quad \text{E_IFFALSE} \\
 \frac{t_1 \longrightarrow t'_1}{\text{if } t_1 \text{ then } t_2 \text{ else } t_3 \longrightarrow \text{if } t'_1 \text{ then } t_2 \text{ else } t_3} \quad \text{E_IF} \\
 \frac{t_1 \longrightarrow t'_1}{\text{succ } t_1 \longrightarrow \text{succ } t'_1} \quad \text{E_SUCC} \\
 \frac{}{\text{pred } 0 \longrightarrow 0} \quad \text{E_PREDZERO} \\
 \frac{}{\text{pred } (\text{succ } v_1) \longrightarrow v_1} \quad \text{E_PREDSUCC} \\
 \frac{t_1 \longrightarrow t'_1}{\text{pred } t_1 \longrightarrow \text{pred } t'_1} \quad \text{E_PRED} \\
 \frac{}{\text{iszero } 0 \longrightarrow \text{true}} \quad \text{E_ISZEROZERO} \\
 \frac{}{\text{iszero } (\text{succ } v_1) \longrightarrow \text{false}} \quad \text{E_ISZEROSUCC} \\
 \frac{t_1 \longrightarrow t'_1}{\text{iszero } t_1 \longrightarrow \text{iszero } t'_1} \quad \text{E_ISZERO} \\
 \frac{t_1 \longrightarrow t'_1}{t_1 t_2 \longrightarrow t'_1 t_2} \quad \text{E_APP1} \\
 \frac{t_2 \longrightarrow t'_2}{v_1 t_2 \longrightarrow v_1 t'_2} \quad \text{E_APP2} \\
 \frac{}{(\lambda x : T. t_{12}) v_2 \longrightarrow [x \mapsto v_2] t_{12}} \quad \text{E_APPABS} \\
 \frac{t_1 \longrightarrow t'_1}{t_1; t_2 \longrightarrow t'_1; t_2} \quad \text{E_SEQ} \\
 \frac{}{\text{unit}; t_2 \longrightarrow t_2} \quad \text{E_SEQNEXT} \\
 \frac{}{v_1 \text{ as } T \longrightarrow v_1} \quad \text{E_ASCRIBE} \\
 \frac{t_1 \longrightarrow t'_1}{t_1 \text{ as } T \longrightarrow t'_1 \text{ as } T} \quad \text{E_ASCRIBE1}
 \end{array}$$

$$\begin{array}{c}
\frac{}{\text{let } x = v_1 \text{ in } t_2 \longrightarrow [x \mapsto v_1]t_2} \text{E_LETV} \\
\frac{t_1 \longrightarrow t'_1}{\text{let } x = t_1 \text{ in } t_2 \longrightarrow \text{let } x = t'_1 \text{ in } t_2} \text{E_LET} \\
\frac{}{\{v_1, v_2\}.1 \longrightarrow v_1} \text{E_PAIRBETA1} \\
\frac{}{\{v_1, v_2\}.2 \longrightarrow v_2} \text{E_PAIRBETA2} \\
\frac{t_1 \longrightarrow t'_1}{t_1.1 \longrightarrow t'_1.1} \text{E_PROJ1} \\
\frac{t_1 \longrightarrow t'_1}{t_1.2 \longrightarrow t'_1.2} \text{E_PROJ2} \\
\frac{t_1 \longrightarrow t'_1}{\{t_1, t_2\} \longrightarrow \{t'_1, t_2\}} \text{E_PAIR1} \\
\frac{t_2 \longrightarrow t'_2}{\{v_1, t_2\} \longrightarrow \{v_1, t'_2\}} \text{E_PAIR2} \\
\frac{}{\text{case } (\text{inl } v_0) \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow [x_1 \mapsto v_0]t_1} \text{E_CASEINL} \\
\frac{}{\text{case } (\text{inr } v_0) \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow [x_2 \mapsto v_0]t_2} \text{E_CASEINR} \\
\frac{t_0 \longrightarrow t'_0}{\text{case } t_0 \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 \longrightarrow \text{case } t'_0 \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2} \text{E_CASESM} \\
\frac{t_1 \longrightarrow t'_1}{\text{inl } t_1 \longrightarrow \text{inl } t'_1} \text{E_INL} \\
\frac{t_1 \longrightarrow t'_1}{\text{inr } t_1 \longrightarrow \text{inr } t'_1} \text{E_INR} \\
\frac{}{\text{fix } (\lambda x : T_1. t_2) \longrightarrow [x \mapsto (\text{fix } (\lambda x : T_1. t_2))]]t_2} \text{E_FIXBETA} \\
\frac{t_1 \longrightarrow t'_1}{\text{fix } t_1 \longrightarrow \text{fix } t'_1} \text{E_FIX} \\
\frac{j \text{ INDEXES } v_i^{i \in 1..n}}{\{v_i^{i \in 1..n}\}.j \longrightarrow v_j} \text{E_PROJTUPLE} \\
\frac{t_1 \longrightarrow t'_1}{t_1.i \longrightarrow t'_1.i} \text{E_PROJTP} \\
\frac{t \longrightarrow t'}{\{v_i^{i \in 1..m}, t, t_j^{j \in 1..n}\} \longrightarrow \{v_i^{i \in 1..m}, t', t_j^{j \in 1..n}\}} \text{E_TUPLE} \\
\frac{j \text{ INDEXES } t_i^{i \in 1..n}}{\text{case } (\langle l_j = v \rangle \text{ as } T) \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} \longrightarrow [x_j \mapsto v]t_j} \text{E_CASEVARIANT} \\
\frac{t \longrightarrow t'}{\text{case } t \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} \longrightarrow \text{case } t' \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n}} \text{E_CASE} \\
\frac{t_i \longrightarrow t'_i}{\langle l_i = t_i \rangle \text{ as } T \longrightarrow \langle l_i = t'_i \rangle \text{ as } T} \text{E_VARIANT}
\end{array}$$

$\overline{\Gamma \vdash \text{true} : \text{Bool}}$	T_TRUE
$\overline{\Gamma \vdash \text{false} : \text{Bool}}$	T_FALSE
$\frac{\Gamma \vdash t_1 : \text{Bool} \quad \Gamma \vdash t_2 : T \quad \Gamma \vdash t_3 : T}{\Gamma \vdash \text{if } t_1 \text{ then } t_2 \text{ else } t_3 : T}$	T_IF
$\overline{\Gamma \vdash 0 : \text{Nat}}$	T_ZERO
$\frac{\Gamma \vdash t_1 : \text{Nat}}{\Gamma \vdash \text{succ } t_1 : \text{Nat}}$	T_SUCC
$\frac{\Gamma \vdash t_1 : \text{Nat}}{\Gamma \vdash \text{pred } t_1 : \text{Nat}}$	T_PRED
$\frac{\Gamma \vdash t_1 : \text{Nat}}{\Gamma \vdash \text{iszero } t_1 : \text{Bool}}$	T_ISZERO
$\frac{x : T \in \Gamma}{\Gamma \vdash x : T}$	T_VAR
$\frac{\Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \lambda x : T_1. t_2 : T_1 \rightarrow T_2}$	T_ABS
$\frac{\Gamma \vdash t_1 : T_{11} \rightarrow T_{12} \quad \Gamma \vdash t_2 : T_{11}}{\Gamma \vdash t_1 t_2 : T_{12}}$	T_APP
$\overline{\Gamma \vdash \text{unit} : \text{Unit}}$	T_UNIT
$\frac{\Gamma \vdash t_1 : \text{Unit} \quad \Gamma \vdash t_2 : T_2}{\Gamma \vdash t_1; t_2 : T_2}$	T_SEQ
$\frac{\Gamma \vdash t_1 : T}{\Gamma \vdash t_1 \text{ as } T : T}$	$T_ASCRIBE$
$\frac{\Gamma \vdash t_1 : T_1 \quad \Gamma, x : T_1 \vdash t_2 : T_2}{\Gamma \vdash \text{let } x = t_1 \text{ in } t_2 : T_2}$	T_LET
$\frac{\Gamma \vdash t_1 : T_1 \quad \Gamma \vdash t_2 : T_2}{\Gamma \vdash \{t_1, t_2\} : T_1 \times T_2}$	T_PAIR
$\frac{\Gamma \vdash t_1 : T_1 \times T_2}{\Gamma \vdash t_1.1 : T_1}$	T_PROJ1
$\frac{\Gamma \vdash t_1 : T_1 \times T_2}{\Gamma \vdash t_1.2 : T_2}$	T_PROJ2
$\frac{\Gamma \vdash t_1 : T_1}{\Gamma \vdash \text{inl } t_1 : T_1 + T_2}$	T_INL
$\frac{\Gamma \vdash t_1 : T_2}{\Gamma \vdash \text{inr } t_1 : T_1 + T_2}$	T_INR

$$\begin{array}{c}
\frac{\Gamma \vdash t_0 : T_1 + T_2 \quad \Gamma, x_1 : T_1 \vdash t_1 : T \quad \Gamma, x_2 : T_2 \vdash t_2 : T}{\Gamma \vdash \text{case } t_0 \text{ of } \text{inl } x_1 \Rightarrow t_1 \mid \text{inr } x_2 \Rightarrow t_2 : T} \quad \text{T_CASESM} \\
\\
\frac{\Gamma \vdash t_1 : T_1 \rightarrow T_1}{\Gamma \vdash \text{fix } t_1 : T_1} \quad \text{T_FIX} \\
\\
\frac{\Gamma \vdash t_i : T_i^{i \in 1..n}}{\Gamma \vdash \{t_i^{i \in 1..n}\} : \{T_i^{i \in 1..n}\}} \quad \text{T_TUPLE} \\
\\
\frac{\Gamma \vdash t_1 : \{T_i^{i \in 1..n}\} \quad j \text{ INDEXES } T_i^{i \in 1..n}}{\Gamma \vdash t_1.j : T_j} \quad \text{T_PROJTP} \\
\\
\frac{\Gamma \vdash t : T_j \quad j \text{ INDEXES } T_i^{i \in 1..n}}{\Gamma \vdash \langle l_j = t \rangle \text{ as } \langle l_i : T_i^{i \in 1..n} \rangle : \langle l_i : T_i^{i \in 1..n} \rangle} \quad \text{T_VARIANT} \\
\\
\frac{\Gamma \vdash t : \langle l_i : T_i^{i \in 1..n} \rangle \quad \Gamma, x_i : T_i \vdash t_i : T^{i \in 1..n} \quad \text{DISTINCT } l_i^{i \in 1..n}}{\Gamma \vdash \text{case } t \text{ of } \langle l_i = x_i \rangle \Rightarrow t_i^{i \in 1..n} : T} \quad \text{T_CASE}
\end{array}$$

Definition rules: 63 good 0 bad
 Definition rule clauses: 119 good 0 bad